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Microsatellite DNA polymorphism and heterozygosity in the malaria vector mosquito *Anopheles funestus* (Diptera: Culicidae) in east and southern Africa

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Abstract

There has been an increase in malaria cases in southern African countries in recent years due to the presence of populations of *Anopheles funestus* that are resistant to the pyrethroid class of insecticides. Since *A. funestus* is one of the major African malaria vectors, knowledge of its genetic structure will benefit control strategies, such as the management of insecticide resistance, by allowing predictions to be made of possible spread of the resistance. This study uses microsatellite DNA markers to analyze samples from five countries in east (Kenya and Uganda), central (Malawi) and southern (South Africa and Mozambique) Africa. There were deviations from Hardy–Weinberg expectations for some loci in all population samples but this was probably due to the presence of null alleles. High levels of genetic diversity were observed (mean alleles per locus = 6.5–10; unbiased H = 0.23–0.89). Low differentiation was observed between Kenya and Uganda (average F_{ST} = 0.002, R_{ST} = 0.0001) and between Mozambique and South Africa (F_{ST} = 0.0004, R_{ST} = 0.02), contrary to high differentiation among the central and southern Africa samples (average F_{ST} = 0.023, R_{ST} = 0.027). High differentiation was measured across the region (mean F_{ST} = 0.04, R_{ST} = 0.08), east versus Malawi (F_{ST} = 0.067, R_{ST} = 0.089) or southern Africa populations (F_{ST} = 0.068, R_{ST} = 0.15). A test of isolation by distance along the east–central–south transect gave evidence (R^2 = 0.50, $P < 0.001$) that geographic distance limits gene flow in *A. funestus*.
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Keywords: *Anopheles funestus*; Malaria; Microsatellite DNA; Heterozygosity

1. Introduction

Anopheles funestus Giles is one of the most important African vectors of malaria and is widely distributed throughout much of the continent. It belongs to a group of nine morphologically similar species (Gillies and Coetzee, 1987) of which only *A. funestus*

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and *A. rivulorum* Leeson (Gillies and Coetzee, 1987; Wilkes et al., 1996) play a role in malaria transmission. *A. funestus* breeds in permanent, well vegetated water bodies such as swamps and slow-moving streams (Gillies and De Meillon, 1968) and consequently is less affected by the vagaries of rainfall, maintaining malaria transmission throughout the year. It is highly anthropophilic and endophilic, making it susceptible to malaria vector control strategies such as indoor residual insecticide house spraying or the use of insecticide treated bed nets.

A. funestus was the major vector of malaria in South Africa prior to the introduction of DDT spraying of houses from the 1950s onwards. This strategy effectively eradicated *A. funestus* from this country. However, in 1996 pyrethroids were introduced in South Africa to replace DDT (Govere et al., 2002) and *A. funestus* returned to the northern areas of Kwazulu/Natal Province where it borders on southern Mozambique, causing a malaria epidemic in 2000 (Hargreaves et al., 2000; Govere et al., 2002). Resistance to pyrethroid insecticides has been demonstrated in Kwazulu/Natal as well as in southern Mozambique (Hargreaves et al., 2000; Brooke et al., 2001). There is obvious concern regarding the potential spread of this resistance and the impact it will have on malaria control programs in the region.

The spread of genes through populations can be predicted by studying the population genetic structure of an organism and estimating gene flow between two or more sub-populations. Various genetic tools are available for such studies and have been used extensively on the other major African malaria vector mosquitoes, *A. gambiae* Giles and *A. arabiensis* Patton (Lehmann et al., 1996; Lehmann et al., 1997; Lanzaro et al., 1998; Donnelly et al., 2001). However, very few reports on population differentiation of *A. funestus* have been published. Cytological studies in Burkina Faso reported two coexisting taxonomic units of *A. funestus* that differ in the degree of chromosomal polymorphism, host preference and resting habits (Costantini et al., 1999). Likewise, significant chromosomal differentiation between west and coastal Kenya has been reported in *A. funestus* (Kamau et al., 2002).

In the present study, the population structure of samples of *A. funestus* from five African countries was examined using microsatellite DNA markers to determine the degree of differentiation between geographically widely separated populations. Comparisons were made between insecticide resistant populations from Mozambique and South Africa and those from Malawi, Kenya and Uganda where no pyrethroid resistance is known to occur.

2. Materials and methods

2.1. Mosquito sampling, DNA extractions and identification

The samples used in this study came from preserved collections in the National Institute for Communicable Diseases, Johannesburg, that had been deposited by collaborators working on different projects. Mosquito samples originated from: the Ndumu area of Kwazulu/Natal Province (27°02′S, 32°19′E) where it borders on southern Mozambique, South Africa (1999); Beluluane (25°53′S, 32°52′E) villages around the MOZAL Aluminium Smelter Plant, Maputo, Mozambique (2000); villages close to Blantyre (15°46′S, 35°00′E) in Malawi (2001); Kisumu (0°03′S, 34°47′E), western Kenya (2000); and Jinja (0°27′N, 33°14′E), Uganda (2001). See Table 2 for geographic distances between these localities.

DNA was extracted from each individual mosquito (Collins et al., 1987) and the resulting DNA pellet re-suspended in 100 ul 1× TE buffer. Field-collected mosquitoes were identified as *A. funestus* using the polymerase chain reaction (PCR) assay (Koekemoer et al., 2002).

2.2. Microsatellite analysis

Initially 12 out of 22 microsatellite loci isolated from *A. funestus* (Sinkins et al., 2000) were tested for this work. They were screened to identify the most suitable pairs for multiplex PCR amplification. Only a multiplex of three pairs (six loci) that produced consistent PCR amplification and polymorphic products were used for this study. Loci used and GeneBank accession number in parentheses are as follows: AF6 (AF171036), AF8 (AF171038), AF5 (AF171035), AF17 (AF171047), AF19 (AF171049) and AF20 (AF171050). The loci: AF2, AF3, AF7, AF14, AF16 and AF18 were not used because of lack of consistent amplification in any pair combination attempted

for multiplex PCR. Microsatellite markers were PCR amplified from 10 ng (1 ul) of *A. funestus* DNA. Each 25 ul reaction consisted of 1.5× Buffer (10 mM Tris pH 8.3, 50 mM KCl), 2.0 mM $MgCl_2$, 0.3 mM dNTPs; 0.2 mM Spermidine; 0.5 U of recombinant *Taq* DNA polymerase (TAKARA Biomedicals) and 12 pmol of each primer. The forward primer of each locus was labeled with a fluorescent dye (FAM, TET, or HEX) compatible with ABI PRISM (Perkin-Elmer, Norwalk, CT) electrophoresis. DNA amplifications were completed in an OMNIGENE HYBAID thermal cycler (MJ Research, Watertown, MA). A 2 min denaturation step at 95 °C was followed by 35 cycles of 30 s at 95 °C, 30 s at 56 °C and 35 s at 72 °C. A final extension at 72 °C was extended to 30 min to alleviate problems associated with addition of non-templated nucleotide (dA) to the PCR products. These conditions were optimized for a multiplex PCR of two loci, the choice of which was based on significant differences in product size. The multiplex pairs were as follows: AF6 and AF8, AF5 and AF17, AF19 and AF20. The products of multiplex PCR were combined for genotyping. Microsatellite allele sizing was performed on an ABI 377 machine (Perkin-Elmer) using standard protocols.

2.3. Analysis of population genetic diversity

Genetic diversity was quantified in terms of heterozygosity, number of alleles per locus and allele frequencies observed in each population sample. Departure from Hardy–Weinberg Equilibrium (HWE) and genotypic homogeneity was tested using Fisher's exact test to determine the level of differences in allele frequencies existing among individuals in a population sample performed with GENEPOP 3.0 (Raymond and Rousset, 1995). Estimates of heterozygosity H_O (observed direct count), H_E (expected Nei's 1978 unbiased) and H_S (expected under HWE) were calculated using TFPGA version 1.3 (Miller, 1997). ANOVA analysis of average heterozygosity was performed to estimate the level of gene diversity among population samples (Weir, 1990). Variations in heterozygosity were analysed with populations, interactions of loci and individuals as factors. All factors were treated as random effects except loci.

Population differentiation was examined by fixation index *F*-statistics (Wright, 1978) and an analogous estimator for microsatellite data R_{ST} (Slatkin, 1995). Estimates of F_{ST} were computed according to (Weir and Cockerham, 1984) using FSTAT (Goudet, 1995). Allele lengths were converted into repeat numbers (Lehmann et al., 1996) and estimates of R_{ST} performed using R_{ST} CALC (Goodman, 1997). R_{ST} CALC corrects for sources of bias from differences in variance between loci and differences in sample size between populations. Genotypic permutation tests were used to evaluate if F_{ST} and R_{ST} estimates differed significantly from zero. The F_{ST} and R_{ST} calculations based on the stepwise mutation model (SMM) and the infinite alleles model (IAM) cover opposite spectrums of theories of microsatellite evolution, therefore providing robust estimates of genetic differentiation.

Relationships among population samples revealed by F_{ST} were assessed using un-weighted pair-group method analysis; UPGMA (Swofford and Olsen, 1990) performed with TFPGA version 1.3 (Miller, 1997). The tree was built based on unbiased genetic distance, with correction for small sample size (Nei, 1978).

The extent of population structure was examined using analysis of molecular variance AMOVA (Excoffier et al., 1992) implemented in ARLEQUIN (Schneider et al., 2000). The AMOVA calculates Φ_{ST}, analogous to Wright (1965) *F*-statistics, an indication of population subdivision. This analysis was carried out for three comparisons: (a) all populations were treated as one group to explore the level of variation between and within them; (b) between east Africa (Kenya and Uganda) and southern Africa excluding Malawi; and (c) between Malawi and either east Africa or Mozambique and South Africa. The standard variance and fixation index (Φ_{ST}) measures can be used to infer the degree of population subdivision. The significance of the resulting fixation index F_{ST} and variance component were tested with 10,100 permutations.

Isolation by distance, a correlation between genetic and geographical distance was tested by the regression of $F_{ST}/1 - F_{ST}$ on the natural logarithm (ln) of geographical distance (Rousset, 1997) performed by Mantel test in TFPGA (Miller, 1997). A similar test was conducted on R_{ST} estimates and significance level obtained after a 5000 permutations. Straight-line geographic distances were determined by using World Atlas in ArcView GIS 3.2 software by ESRI.

Before calculating *F*-statistics, we determined whether the variation at all loci appeared to be neutral so that genetic drift could be inferred as the primary force affecting them. This can be assumed if all loci have similar values of F_{ST} (Slatkin, 1985). A crude test for this is to see whether the confidence intervals (CI) of F_{ST} calculated over populations, overlap or lie between the CI of F_{ST} calculated over loci (Goudet et al., 1994). If these overlap, the loci can be assumed to behave in a neutral manner, loci not meeting this criterion should be omitted from the analysis. FSTAT calculates these CI using jack knife procedure (Weir, 1990; Goudet, 1995).

3. Results

The microsatellite markers used in this study have not been physically mapped to the polytene chromosomes of *A. funestus*. It is unknown, therefore, whether any of them are situated within or close to the breakpoints of paracentric inversions. If they are, this may well alter the analysis and interpretation of the data since one cannot assume that the markers are neutral if they are linked to polymorphic chromosomal inversions (Krzywinski and Besansky, 2003). However, over population samples confidence intervals of F_{ST} for each locus overlapped or lay between over loci CI (data not shown) and therefore all six loci were used for estimation of genetic differentiation.

Moderate to high levels of polymorphism were found in *A. funestus* samples from the five African countries (Table 1). The Kenya and Uganda samples are referred to as east Africa populations, Malawi as central Africa and Mozambique and South Africa as southern Africa populations. In total, the average number of alleles per locus was 7.4 (range 4–13) and the average heterozygosity was 0.7 (range 0.42–0.86). The east African samples showed an average of 8.5 (range 5–11) alleles per locus with average heterozygosity of 0.76 (range 0.65–0.82). The Malawi population had an average of 6.7 alleles (range 4–10) and average heterozygosity of 0.704 (range 0.60–0.82). Likewise samples from southern Africa had a lower average of 6.6 (range 4–13) alleles per locus with average heterozygosity of 0.66 (range 0.42–0.86). There was low DNA polymorphism at loci AF6 and AF8 for the Malawi, Mozambique and South Africa samples (average H_o for AF6 = 0.3773: AF8 = 0.4232) compared with east Africa (average H_o for AF6 = 0.5651: AF8 = 0.7219). While the average number of alleles differed between east and central/southern Africa, some loci showed varied distribution patterns among samples (Fig. 1), no further analysis on number of alleles was performed because they are sample size dependent and the possibility of null alleles skewing the data.

3.1. Test for Hardy–Weinberg equilibrium and linkage disequilibrium

Hardy–Weinberg predictions were significantly rejected when the pooled samples were analyzed as a single population. With the exception of locus AF6, the rest of the loci were in HWE for most of the samples (Table 1). Significant heterozygote deficits were detected in 5 out of 18 tests ($P < 0.001$) for the Malawi and two population samples from southern Africa. These were clustered at loci AF6 and AF8. In east Africa, 3 out of 12 ($P < 0.001$) tests indicated significant heterozygote deficits at AF6 and AF19 (Table 1). Heterozygote deficit may be due to the Wahlund effect (pooling of separate gene pools), inbreeding and or null alleles. Similar studies in *A. gambiae* and *A. arabiensis* have shown evidence of deviation from HWE (Lanzaro et al., 1998; Donnelly et al., 1999) and this might indicate a common phenomenon in microsatellite polymorphism, which is also reflected in *A. funestus*.

Analysis of linkage disequilibrium was carried out to confirm whether the deviations from HWE within-samples were not due to Wahlund effect, inbreeding, locus specific events such as null alleles or linkage with genes under selection pressure. Exact tests for linkage disequilibrium within each sample resulted in two significant values ($P < 0.01$) out of 45 comparisons (4.4%) in Malawi, Mozambique and South Africa populations compared to one ($P < 0.01$) out of 30 comparisons (3.3%) for east Africa. It can be concluded that there is no linkage disequilibrium within the *A. funestus* populations sampled. Unlike the Wahlund effect and inbreeding, which affect the entire genome, the observed deficits were clustered on locus AF6 (Table 1) and therefore it is unlikely that population subdivision is due to Wahlund or inbreeding effects. Indeed, there was repeated PCR

Table 1
Sample size (N), total number of alleles (A_T), heterozygosity expected under HWE (H_S), observed direct count (H_O) and expected Nei's 1978 unbiased (H_E) heterozygosity by locus for samples of *Anopheles funestus* from five African countries

		Mozambique	S. Africa	Malawi	Kenya	Uganda
AF6	N	42	33	34	42	34
	A_T	5	6	7	11	10
	H_E	0.7128	0.7296	0.7643	0.8313	0.8121
	H_O	0.2381	0.3939	0.5000	0.5714	0.5588
	H_S	0.647	0.718	0.753	0.821	0.800
AF8	N	47	33	36	43	34
	A_T	7	6	7	10	10
	H_E	0.7051	**0.6731**	0.7692	**0.7953**	**0.8481**
	H_O	0.3404	**0.4848**	0.4444	**0.7674**	**0.6765**
	H_S	0.655	0.663	0.758	0.786	0.836
AF5	N	48	33	38	43	34
	A_T	5	5	5	5	5
	H_E	**0.5612**	**0.5543**	**0.6021**	**0.6312**	**0.6795**
	H_O	**0.6042**	**0.6364**	**0.7368**	**0.6279**	**0.5882**
	H_S	0.526	0.546	0.594	0.624	0.669
AF17	N	45	33	38	42	33
	A_T	11	13	10	10	8
	H_E	**0.8567**	**0.8774**	**0.8242**	**0.8333**	**0.7963**
	H_O	**0.8222**	**0.7273**	**0.8684**	**0.5714**	**0.6364**
	H_S	0.841	0.864	0.813	0.823	0.784
AF19	N	46	32	35	41	33
	A_T	6	7	7	8	9
	H_E	**0.4345**	**0.6299**	**0.6451**	0.6736	**0.7599**
	H_O	**0.4186**	**0.6562**	**0.5714**	0.2927	**0.6666**
	H_S	0.421	0.620	0.636	0.665	0.748
AF20	N	46	32	37	43	34
	A_T	4	4	4	7	9
	H_E	**0.5509**	**0.6096**	**0.6205**	**0.7499**	**0.8046**
	H_O	**0.5217**	**0.5312**	**0.4594**	**0.5814**	**0.6765**
	H_S	0.568	0.600	0.612	0.741	0.793
Means ± S.E.	A_T	6.3 ± 0.93	6.8 ± 0.19	6.7 ± 0.77	8.5 ± 0.84	8.5 ± 0.69
	H_E	0.6386 ± 0.05	0.6789 ± 0.04	0.7042 ± 0.03	0.7524 ± 0.03	0.7834 ± 0.06
	H_O	0.4909 ± 0.07	0.5711 ± 0.05	0.5967 ± 0.06	0.5687 ± 0.06	0.6338 ± 0.02
	H_S	0.609	0.669	0.694	0.744	0.772

Loci at Hardy–Weinberg equilibrium are in bold.

amplification failure for locus AF6, suggesting that they were homozygous for null alleles.

3.2. Genetic diversity

The gene diversity was assessed to determine differences in average heterozygosity between east, central and southern African population samples. This analysis was significant ($F = 5.8$, $P < 0.0001$) suggesting differences in gene diversity along this north–south transect. Overall, diversity was significant at locus AF6 ($F = 3.3$; $P = 0.013$), AF8 ($F = 5.7$; $P < 0.0003$), AF17 ($F = 3.23$; $P = 0.014$) and AF19 ($F = 4.5$; $P = 0.002$). Furthermore, the observed average heterozygosity varied among loci and location of samples (ANOVA for hetorezygosity: east Africa; $F = 2.4$, $P = 0.036$, Malawi; $F = 4.9$, $P = 0.0003$, southern Africa (Mozambique and S. Africa); $F = 9.34$, $P < 0.001$). In east Africa, gene diversity was highly significant at locus AF19 ($P = 0.008$) compared with

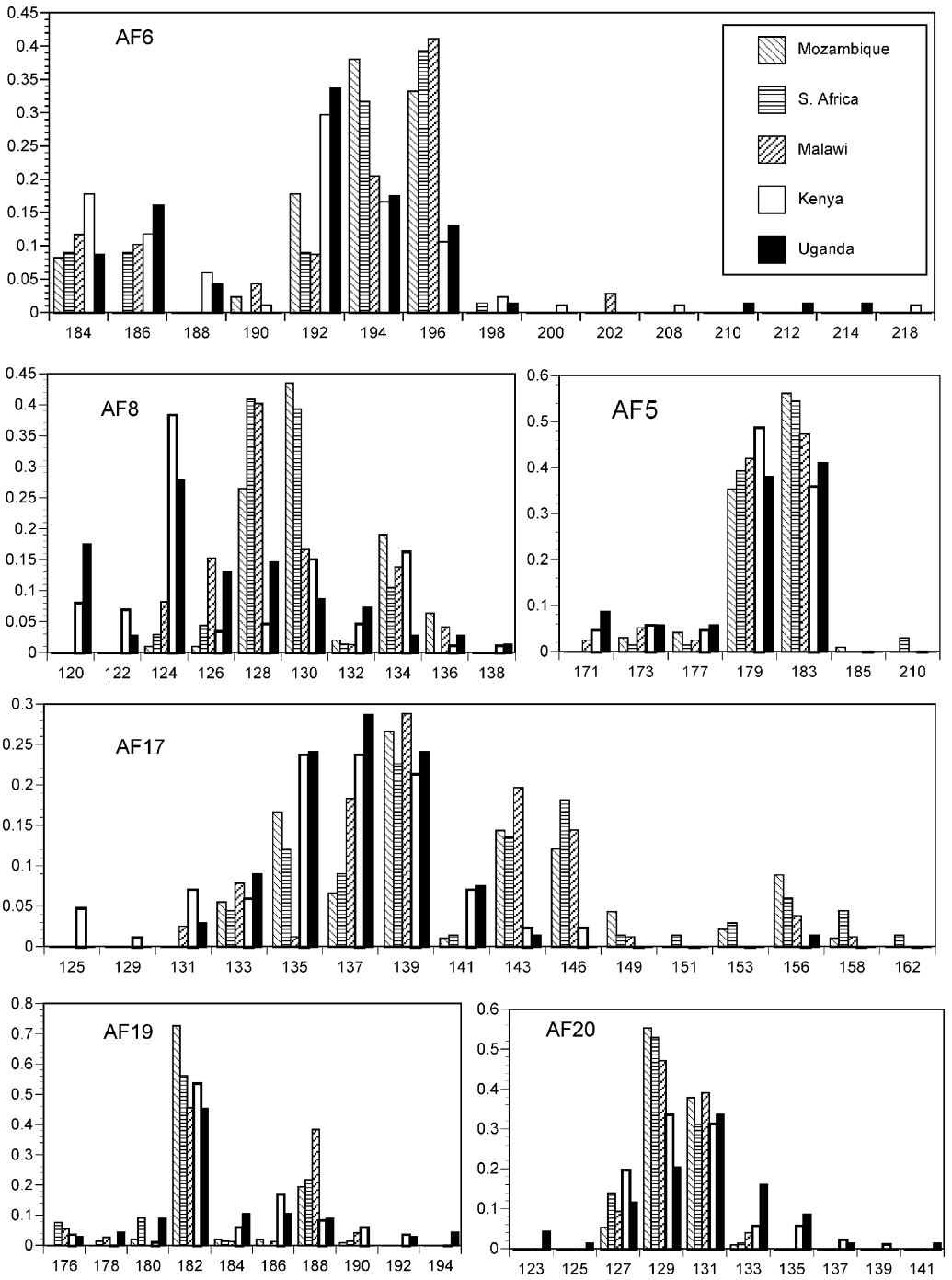


Fig. 1. Allele frequency distribution in each sample of *A. funestus* for each microsatellite locus. Alleles are denoted by their estimated sizes (bp).

Table 2
Genetic differentiation estimates for each sample pair

	Mozambique	S. Africa	Malawi	Kenya	Uganda
Mozambique		0.0003 [139] (*0.148*)	**0.0295** [1144] (*0.001*)	**0.0679** [2872] (*0.0001*)	**0.0763** [2915] (*0.0001*)
S. Africa	0.0136 (*0.202*)		0.0060[a] [1275] (*0.088*)	**0.0645** [2999] (*0.0001*)	**0.0649** [3047] (*0.0001*)
Malawi	**0.0276** (*0.017*)	0.0367 (*0.029*)		**0.0597** [1742] (*0.0001*)	**0.0544** [1881] (*0.0001*)
Kenya	**0.1650** (*0.000*)	**0.1462** (*0.000*)	**0.0677** (*0.000*)		0.0020 [188] (*0.0457*)
Uganda	**0.1639** (*0.000*)	**0.1392** (*0.000*)	**0.0730** (*0.000*)	−0.0006 (*0.524*)	

Below diagonal are R_{ST} and above diagonal are F_{ST} (Weir and Cockerham, 1984) estimates. The significance values are in italics within the brackets. Genetic distance values in bold letters indicate significance after adjustment by the sequential Bonferroni procedure. Pairwise geographical distances in km are in squared brackets above the diagonal.

[a] Exact *G*-test $P = 0.005$, whereas $P = 0.08$ after sequential Bonferroni correction.

loci AF5 ($P = 0.04$) and AF17 ($P = 0.0001$) for the Malawi, and AF6 ($P < 0.001$), AF5 ($P = 0.006$) and AF17 ($P < 0.001$) for the southern Africa populations.

3.3. Genetic differentiation

Overall F_{ST}/R_{ST} were significant for most loci, average F_{ST} of 0.044, ranging from 0.009 at locus AF5 to 0.098 at locus AF8 and average R_{ST} of 0.088, ranging from 0.012 at locus AF19 to 0.21 at locus AF17 (data not shown), indicating that differentiation is genome wide. The rare alleles (frequencies <5%) were pooled with alleles of similar size and this effect on differentiation assessed. Pooling of rare alleles insignificantly increased these overall estimates: average F_{ST} of 0.074 (range: 0.036–0.164), R_{ST} of 0.109 (range: 0.012–0.220).

Estimates of F_{ST} and R_{ST} varied between loci and region with a similar pattern of differentiation between east, central and southern Africa samples (Tables 2 and 3). Pairwise comparison of all populations (Table 2) or sample by sample comparison (Table 3) did not reveal significant differentiation between either Mozambique versus South Africa ($F_{ST} = 0.0003$, $P = 0.15$; R_{ST} 0.013, $P = 0.2$) or Kenya versus Uganda ($F_{ST} = 0.002$, $P = 0.046$; $R_{ST} = -0.0006$, $P = 0.52$). These results were consistent for both estimates and analysis involving pooling of rare alleles. Where genetic distance is negative it means that the within-sample variance in allele size is bigger than the among-sample variance (Goodman, 1997). Thus, the genetic variation is within-samples rather than between, which means no differentiation and the populations are essentially panmictic, as it applies to Kenya versus Uganda and Mozambique versus South Africa (Tables 2 and 3).

Genetic distance analysis revealed significant variation between east and either central or southern African ($F_{ST} = 0.06$, $P < 0.001$) *A. funestus*. All

Table 3
Differentiation per locus of populations from east (Kenya and Uganda) central (Malawi) and southern (Mozambique and S. Africa) Africa

	Kenya vs. Uganda		S. Afri vs. Mozambique		Malawi vs. east		Malawi vs. southern		East vs. southern	
	F_{ST}	R_{ST}	F_{ST}	R_{ST}	F_{ST}	R_{ST}	F_{ST}	R_{ST}	F_{ST}	R_{ST}
AF6	−0.009	−0.0008	−0.0045	−0.0094	0.0713	0.0168	0.0068	−0.0038	0.0738	0.0556
AF8	0.024	0.0078	0.0069	0.1142	0.0954	0.1423	0.0461	0.0612	0.1442	0.2964
AF5	−0.002	−0.0125	−0.008	0.0107	−0.0011	0.0163	0.0014	0.0379	0.0269	0.1045
AF17	−0.012	0.0134	0.008	−0.0005	0.0573	0.2658	0.0123	0.02005	0.0441	0.3338
AF19	0.0004	−0.0065	0.0247	0.0053	0.0654	0.0005	0.0558	0.0467	0.0407	0.0112
AF20	0.009	−0.0001	−0.0034	0.0111	0.0260	0.0233	−0.0051	−0.0016	0.0564	0.0483
Overall	0.002ns	0.0001ns	0.0004ns	0.0217ns	0.0674**	0.0887**	0.0238**	0.0271*	0.0682***	0.1504***
P-values	0.05	0.49	0.15	0.07	0.001	0.0001	0.001	0.007	0.001	0.0000

The F_{ST} and R_{ST} estimates were calculated using FSTAT and R_{ST} CALC respectively. Pairwise significance after standard Bonferroni corrections; ***$P < 0.001$; **$P < 0.01$; *$P < 0.05$ and NS stands for non-significant.

pairwise comparisons showed significant population differentiation between these three groups (Table 3). Genetic distance between Malawi and either southern Africa (mean $F_{ST} = 0.024$, $R_{ST} = 0.027$) or east Africa (mean $F_{ST} = 0.067$, $R_{ST} = 0.089$) was highly significant, indicating the uniqueness of the Malawi sample (Table 3). The differentiation of the Malawi population from the two southern African populations increased after pooling of rare alleles: Malawi versus Mozambique, $F_{ST} = 0.029$ (Exact *G*-test $P = 0.005$, $P = 0.001$ after sequential Bonferroni correction) before pooling and $F_{ST} = 0.034$ (EGt $P = 0.005$, $P = 0.001$ sBC) after pooling of rare alleles; Malawi versus S. Africa, $F_{ST} = 0.006$ (EGt $P = 0.005$, $P = 0.08$ sBC) before pooling and $F_{ST} = 0.009$ (EGt $P = 0.005$, $P = 0.001$ after sBC) after pooling rare alleles. These results suggest that rare alleles may be responsible for the observed genetic differentiation, at least for Malawi and the southern African populations.

The extent of population subdivision was further investigated by AMOVA analyses which revealed significant population structure with high Φ_{ST} for all comparisons (Table 4). At least 23% of variance was accounted for within population in all comparisons.

A cluster analysis based on mean pairwise F_{ST} to assess the relationship among population samples is as shown in Fig. 2. This analysis provided a good fit for clustering along the east, central and southern Africa transect. The cluster of central and southern African samples was significantly supported (83%) and the branching within this cluster modestly supported (50%), indicating subdivision and providing further evidence of population structure as suggested by previous analyses of the data (Tables 2–4).

3.4. Isolation by distance

Tests of isolation by distance showed a positive correlation between genetic and geographic distances along a north–south transect. The genetic and geographical distances matrix used for this analysis is summarized in Table 2. The test, based on $F_{ST}/1-F_{ST}$ against the natural logarithm of geographical distance, was significant before ($R^2 = 0.50$, $P = 0.0002$) and even after pooling of rare alleles with frequencies less than 5% ($R^2 = 0.46$, $P = 0.009$). Likewise calculation based on R_{ST} estimates gave similar results ($R^2 = 0.65$, $P = 0.014$).

4. Discussion

This work is an early report of microsatellite DNA polymorphism for studying the population structure of *A. funestus*. Significant genetic differentiation ($F_{ST} = 0.044$, $R_{ST} = 0.088$; $P < 0.001$) was found

Table 4
Analysis of molecular variance within and among five populations of *An funestus*

Source of variation	Variance component	Total (%)	Φ_{ST}	*P*
All five populations in one group				
Among populations	0.0919	4.05	0.0404	<0.0001
Within populations	0.5302	23.34		
Kenya and Uganda vs. Mozambique and S. Africa				
Among groups	0.1446	6.22	0.06216	<0.0001
Within groups	0.0028	0.12		
Malawi vs. Mozambique and S. Africa				
Among groups	0.0371	1.76	0.0176	<0.001
Within groups	0.0022	0.11		
Malawi vs. Kenya and Uganda				
Among groups	0.1164	4.83	0.0483	<0.0001
Within groups	0.0039	0.16		
Kenya and Uganda vs. Malawi vs. Mozambique and S. Africa				
Among groups	0.1111	4.84	0.0484	<0.0001
Within groups	0.0029	0.13		

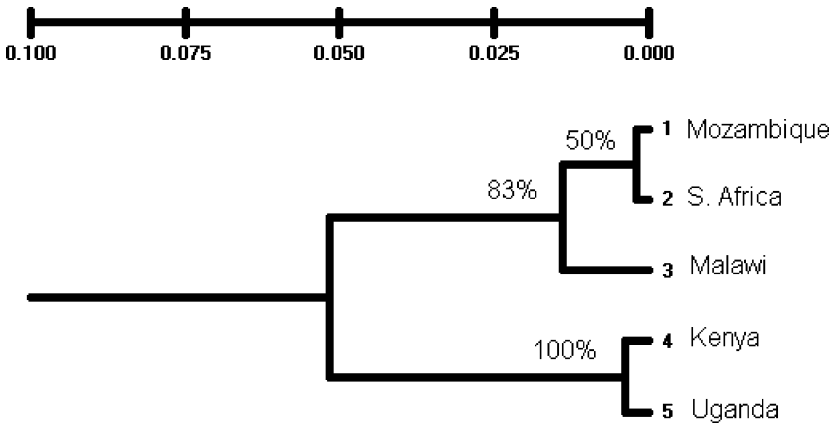


Fig. 2. A UPGMA phenogram illustrating relationship among *A. funestus* populations sampled from southern (South Africa, Mozambique and Malawi) and east Africa (Kenya and Uganda) regions. This tree was constructed based on Nei's unbiased genetic distance. The bootstrap (1000 permutations) support for the branching is indicated in percentages.

between the five samples from east, central and southern African countries (Tables 1–4).

There was no evidence of linkage disequilibrium suggesting that samples were drawn from homogeneous, randomly mating populations. The heterozygote deficits observed at locus AF6 are probably attributable to null alleles since there was repeated amplification failure for this locus.

Cluster analysis was used to evaluate genetic distance statistics with respect to geographic structuring. This analysis resolved regional (east versus central versus south) structuring of *A. funestus* populations with the level of support for clustering between 50 and 83% (Fig. 2). The east African samples were clearly different from the others while the results provided evidence of structuring between central and southern African populations.

In east Africa, our study compared *A. funestus* from Kisumu, Kenya and Jinja, Uganda, 200 km apart. The two areas are located close to Lake Victoria, west of the Rift Valley. Microsatellite polymorphism revealed reduced genetic differentiation and apparent panmixia. These results are similar to a recent study on microsatellite DNA polymorphism in *A. funestus* that showed insignificant differentiation between two sites (Udhoro and Mbita; 48 km apart) around lake Victoria in western Kenya (Braginets et al., 2003). Similarly, lack of genetic differentiation between two samples of *A. gambiae* located in a comparable ecological setting around Lake Victoria has been observed, leading to a conclusion that the geographical extent of the population in that area is larger than 50 km in diameter (Lehmann et al., 1997).

Tests of isolation by distance gave results consistent with the hypothesis that geographic distance limits gene flow across a north–south transect. Our data showed that differentiation estimated over six loci was high and was significantly correlated with geographic distance (F_{ST}; $R^2 = 0.49$, $P < 0.001$ and R_{ST}; $R^2 = 0.65$, $P = 0.01$).

The fact that Mozambique and South African populations of *A. funestus* are resistant to pyrethroids together with some evidence of resistance to the carbamate Propoxur (Hargreaves et al., 2000; Brooke et al., 2001), led us to investigate the genetic structure of samples from these localities and to compare them with *A. funestus* from other regions from which we had stored samples. The hypothesis was made that selection for resistance would lead to the southern African populations being genetically different to other populations where resistance is not known to have developed. The use of microsatellite DNA markers and statistical analysis for genetic differentiation appear to support this hypothesis. What is surprising is the marked differences seen in the Malawi sample compared to both east and southern samples. At the time that our Mozambican sample was collected, only limited vector control activities were going on in the region of the MOZAL aluminium smelter plant with nothing taking place outside a 1.5 km radius of

the plant nor to the north of Maputo city. There was no reason to suspect that the *A. funestus* population was not continuous through to Malawi which borders on Mozambique about 1200 km north of Maputo, except for the fact that Malawi *A. funestus* were not resistant to pyrethroids (unpublished data). The data presented here, suggesting limited gene flow across those 1200 kms, may mean that the resistance genes will not spread rapidly through southern Africa, although one cannot predict what the effect of selection pressure would be if vector control operations were to be implemented in central and northern Mozambique.

The pattern and amount of heterogeneity detected within a population and across large distances are influenced by a complex array of factors including population sizes, dispersal rates, and demographic changes such as bottlenecks, expansion or selection pressure. There is a need for more detailed studies on population structure, ecology of dispersal and phenotypic adaptation, to understand the functional processes shaping population structure in *A. funestus*.

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